Chapter 6. Conditional Heteroscedasticity Models

ARCH, GARCH, change points, Value-at-Risk

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6.1 Introduction

Motivation. Returns on stocks, interest rates and exchange rates are nearly uncorrelated, yet their size is predictable: large moves cluster together. This chapter models a conditional variance that changes with the past, so that the *volatility* itself becomes the object of inference.

Volatility clustering means that sharp rises and falls come in bursts. Statistically, volatility is a variance: a series is volatile when the conditional variance of the next value, given the present, depends strongly on the current state. Classical models cannot express this. For an AR(1) model $y_t = \phi y_{t-1} + \epsilon_t$ with $\{\epsilon_t\} \sim IID(0, \sigma^2)$,

$$E(y_t \mid y_{t-1}) = \phi y_{t-1}, \qquad \mathrm{Var}(y_t \mid y_{t-1}) = \sigma^2,$$

so the conditional mean moves but the conditional variance is constant. If instead $y_t = 0.1 y_{t-1} + \epsilon_t$ with $\epsilon_t \mid \epsilon_{t-1} \sim N(0,\, 1 + 0.5\epsilon_{t-1}^2)$, a large shock raises the variance of the next one, and the path shows quiet stretches interrupted by turbulent ones (Figure 1).

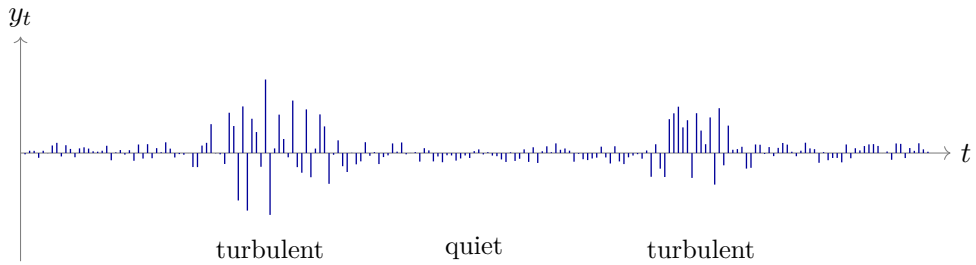


Figure 1: Volatility clustering: the level is unpredictable, but large moves come in bursts.

6.2 ARCH models

Motivation. Engle's ARCH model lets the conditional variance be a function of past squared values. Its simplest version already produces stationary, uncorrelated, heavy-tailed data, and its squares follow an AR model.

Write $\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \dots)$ for the information up to time $t-1$ (the book's I_{t-1}). We wish to model

$$y_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2), \qquad \sigma_t^2 = \sigma_t^2(y_{t-1}, y_{t-2}, \dots), \tag{6.1}$$

a conditional variance that is itself autoregressive in nature; often only finitely many lags $y_{t-1}, \dots, y_{t-q}$ enter.

6.2.1 ARCH(1)

Let $\{\xi_t\} \sim IID\, N(0,1)$ and consider

$$y_t = \sigma_t \xi_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2, \qquad \alpha_0 > 0,\ 0 \le \alpha_1 < 1, \tag{6.2}$$

where ξ_t is independent of σ_t. Since σ_t is determined by y_{t-1} alone,

$$P(y_t \le x \mid y_{t-1} = y) = P(\xi_t \le x/\sqrt{\alpha_0 + \alpha_1 y^2}) = \Phi(x/\sqrt{\alpha_0 + \alpha_1 y^2}), \tag{6.3}$$

with Φ the standard normal distribution function, so $y_t \mid y_{t-1}, y_{t-2}, \cdots \sim N(0, \sigma_t^2)$ and (6.2) is a concrete instance of (6.1): the *ARCH(1) model*. In practice ξ_t is often only assumed $IID(0,1)$.

Iterating $y_t^2 = \sigma_t^2\xi_t^2 = (\alpha_0 + \alpha_1 y_{t-1}^2)\xi_t^2$,

$$y_t^2 = (\alpha_0 + \alpha_1(\alpha_0 + \alpha_1 y_{t-2}^2)\xi_{t-1}^2)\xi_t^2 = \cdots = \alpha_0 \sum_{j=0}^{n} \alpha_1^j \xi_t^2 \cdots \xi_{t-j}^2 + \alpha_1^{n+1}\xi_t^2 \cdots \xi_{t-n}^2 y_{t-n-1}^2, \tag{6.4}$$

and since $\alpha_1^{n+1} \to 0$ this suggests

$$y_t^2 = \alpha_0 \sum_{j=0}^{\infty} \alpha_1^j \xi_t^2 \cdots \xi_{t-j}^2, \qquad \sigma_t^2 = \alpha_0 + \alpha_0\alpha_1 \sum_{j=0}^{\infty} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2, \tag{6.5}$$

that is, the candidate solution

$$y_t = \xi_t\Big(\alpha_0 + \alpha_0\alpha_1 \sum_{j=0}^{\infty} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2\Big)^{1/2}. \tag{6.6}$$

Being a nonlinear function of $\xi_t, \xi_{t-1}, \ldots$, it cannot be captured by a linear ARMA structure.

Proposition (Properties of ARCH(1), [Supplement])

Let $\alpha_0 > 0$, $0 \le \alpha_1 < 1$ and $\{\xi_t\} \sim IID(0,1)$.
(1) The series in (6.6) converges a.s.; $\{y_t\}$ is a strictly stationary solution of (6.2), σ_t^2 is a function of $\xi_{t-1}, \xi_{t-2}, \ldots$, and (σ_t^2, y_t) is independent of $\xi_{t+1}, \xi_{t+2}, \ldots$.
(2) $E(y_t \mid \mathcal{F}_{t-1}) = 0$, $E(y_t^2 \mid \mathcal{F}_{t-1}) = \mathrm{Var}(y_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$, and

$$Ey_t^2 = E\sigma_t^2 = \frac{\alpha_0}{1-\alpha_1}; \tag{6.7}$$

$\{y_t\}$ is a stationary white noise.
(3) If $\xi_t \sim N(0,1)$, then $Ey_t^4 < \infty$ if and only if $3\alpha_1^2 < 1$, and then the kurtosis is

$$\frac{Ey_t^4}{(Ey_t^2)^2} = 3\,\frac{1-\alpha_1^2}{1-3\alpha_1^2} > 3 \quad (\alpha_1 > 0). \tag{6.8}$$

(4) If $Ey_t^4 < \infty$, then $\nu_t = y_t^2 - \sigma_t^2$ is a white noise and

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \nu_t, \tag{6.9}$$

so $\{y_t^2\}$ is an AR(1) process.

Proof.

Step 1. Put $h_t = \alpha_0 + \alpha_0\alpha_1 \sum_{j\ge 0} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2$. The terms are nonnegative, and by independence $E(\xi_{t-1}^2 \cdots \xi_{t-j-1}^2) = 1$, so $Eh_t = \alpha_0 \sum_{j\ge 0} \alpha_1^j = \alpha_0/(1-\alpha_1) < \infty$; hence the series converges a.s. Moreover

$$\alpha_0 + \alpha_1\xi_{t-1}^2 h_{t-1} = \alpha_0 + \alpha_0\alpha_1\xi_{t-1}^2 + \alpha_0 \sum_{j\ge 0} \alpha_1^{j+2} \xi_{t-1}^2\xi_{t-2}^2 \cdots \xi_{t-j-2}^2 = h_t,$$

so with $\sigma_t^2 = h_t$ and $y_t = \sigma_t\xi_t$ we get $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$. Since $(y_t)_t = G(\ldots, \xi_{t-1}, \xi_t)$ for one fixed measurable map G applied to the shifted i.i.d. sequence, $\{y_t\}$ is strictly stationary. The independence claims are immediate from the construction.

Step 2. $\mathcal{F}_{t-1} \subset \sigma(\xi_{t-1}, \xi_{t-2}, \ldots)$, which is independent of ξ_t, while σ_t is $\mathcal{F}_{t-1}$-measurable (it equals $(\alpha_0 + \alpha_1 y_{t-1}^2)^{1/2}$). Hence $E(y_t \mid \mathcal{F}_{t-1}) = \sigma_t E\xi_t = 0$ and $E(y_t^2 \mid \mathcal{F}_{t-1}) = \sigma_t^2 E\xi_t^2 = \sigma_t^2$. Taking expectations, $Ey_t^2 = E\sigma_t^2 = Eh_t = \alpha_0/(1-\alpha_1)$. For $h \ge 1$, $E(y_t y_{t+h}) = E\big(y_t\, E(y_{t+h} \mid \mathcal{F}_{t+h-1})\big) = 0$, so $\{y_t\}$ is white noise with mean 0.

Step 3. Write $y_t^2 = \alpha_0 \sum_{j\ge 0} \alpha_1^j P_j$ with $P_j = \xi_t^2 \cdots \xi_{t-j}^2$. Since $E\xi^4 = 3$, $\|P_j\|_2 = (EP_j^2)^{1/2} = 3^{(j+1)/2}$. If $\sqrt{3}\,\alpha_1 < 1$, Minkowski's inequality gives $\|y_t^2\|_2 \le \alpha_0 \sum_j \alpha_1^j 3^{(j+1)/2} < \infty$, so $Ey_t^4 < \infty$.

Conversely, suppose $Ey_t^4 < \infty$; then $m_4 := E\sigma_t^4$ is finite and, by stationarity, does not depend on t. Using $Ey_{t-1}^4 = 3E\sigma_{t-1}^4$,

$$m_4 = E(\alpha_0 + \alpha_1 y_{t-1}^2)^2 = \alpha_0^2 + 2\alpha_0\alpha_1 \frac{\alpha_0}{1-\alpha_1} + 3\alpha_1^2 m_4, \quad \text{so} \quad (1-3\alpha_1^2)m_4 = \alpha_0^2 \frac{1+\alpha_1}{1-\alpha_1} > 0,$$

forcing $3\alpha_1^2 < 1$. Then $Ey_t^4 = 3m_4$ and

$$\frac{Ey_t^4}{(Ey_t^2)^2} = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)} \cdot \frac{(1-\alpha_1)^2}{\alpha_0^2} = 3\,\frac{1-\alpha_1^2}{1-3\alpha_1^2},$$

which exceeds 3 when $\alpha_1 > 0$ because $1-\alpha_1^2 > 1-3\alpha_1^2$.

Step 4. $\nu_t = \sigma_t^2(\xi_t^2 - 1)$ is $\mathcal{F}_t$-measurable with $E(\nu_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 E(\xi_t^2 - 1) = 0$ and $E\nu_t^2 = E\sigma_t^4\, E(\xi_t^2-1)^2 < \infty$. A square-integrable martingale difference has mean zero, constant variance (by stationarity) and is uncorrelated: for $h \geq 1$, $E(\nu_t\nu_{t+h}) = E(\nu_t E(\nu_{t+h} \mid \mathcal{F}_{t+h-1})) = 0$. Finally $y_t^2 = \sigma_t^2 + \nu_t = \alpha_0 + \alpha_1 y_{t-1}^2 + \nu_t$. ■

Parts (1)–(2) do not use normality. Part (3) shows that the marginal law of an ARCH(1) series is *leptokurtic*: whatever the distribution of ξ_t, y_t has heavier tails than the normal. For example, 10,000 draws from $\sigma_t^2 = 0.2 + 0.8y_{t-1}^2$ produce a histogram far more peaked and heavy-tailed than the fitted normal density.

Definition 6.2.1 (ARCH(q) model)

If $\{y_t\}$ satisfies

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_q y_{t-q}^2, \tag{6.10}$$

with $\alpha_0 > 0$ and $\alpha_i \geq 0$ $(1 \leq i \leq q)$, then $\{y_t\}$ follows an *ARCH(q) model*. If moreover $1 - \alpha_1 z - \cdots - \alpha_q z^q \neq 0$ for all $|z| \leq 1$ (equivalently $\sum_{i=1}^q \alpha_i < 1$), then $\{y_t\}$ is strictly stationary and

$$\mathrm{Var}(y_t) = \frac{\alpha_0}{1 - \sum_{i=1}^q \alpha_i}. \tag{6.11}$$

Proof of the stationarity statement. [Supplement] Let $S = \sum_{i=1}^q \alpha_i$ and $A(z) = \sum_i \alpha_i z^i$.

Step 1: the root condition is $S < 1$. If $S < 1$ then $|A(z)| \leq S < 1$ for $|z| \leq 1$, so $1 - A(z) \neq 0$. If $S \geq 1$, the continuous function A satisfies $A(0) = 0$ and $A(1) = S \geq 1$, so $A(z) = 1$ for some $z \in (0, 1]$.

Step 2: construction. Set $h_t^{(0)} = 0$ and $h_t^{(m+1)} = \alpha_0 + \sum_{i=1}^q \alpha_i \xi_{t-i}^2 h_{t-i}^{(m)}$. By induction each $h_t^{(m)} = g_m(\xi_{t-1}, \xi_{t-2}, \dots)$ for a fixed measurable $g_m \geq 0$, and $h_t^{(m)}$ is nondecreasing in m ($h^{(1)} = \alpha_0 \geq h^{(0)}$, and the recursion is monotone because its coefficients are nonnegative). Since $h_{t-i}^{(m)}$ depends only on $\xi_{t-i-1}, \dots$, it is independent of ξ_{t-i}, so $Eh_t^{(m+1)} = \alpha_0 + S\, Eh_t^{(m)}$ and $Eh_t^{(m)} = \alpha_0(1 + S + \cdots + S^{m-1}) \leq \alpha_0/(1-S)$.

Step 3: limit. Let $\sigma_t^2 = \lim_m h_t^{(m)}$. By monotone convergence $E\sigma_t^2 = \alpha_0/(1-S) < \infty$, so $\sigma_t^2 < \infty$ a.s., and letting $m \to \infty$ in the recursion gives $\sigma_t^2 = \alpha_0 + \sum_i \alpha_i \xi_{t-i}^2 \sigma_{t-i}^2$. With $y_t = \sigma_t \xi_t$ this is (6.10). As in Proposition , $\{y_t\}$ is a fixed function of the shifted i.i.d. sequence, hence strictly stationary, and $\mathrm{Var}(y_t) = E\sigma_t^2 E\xi_t^2 = \alpha_0/(1-S)$, which is (6.11).

Step 4: uniqueness. Let $\tilde{\sigma}_t^2$ be another strictly stationary solution that is a function of $\xi_{t-1}, \xi_{t-2}, \dots$ with $E\tilde{\sigma}_t^2 < \infty$. Then $D_t = \sigma_t^2 - \tilde{\sigma}_t^2$ satisfies $D_t = \sum_i \alpha_i \xi_{t-i}^2 D_{t-i}$, and by independence of ξ_{t-i} and D_{t-i}, $E|D_t| \leq \sum_i \alpha_i E|D_{t-i}| = S\, E|D_0|$. Stationarity gives $(1-S)E|D_0| \leq 0$, so $D_t = 0$ a.s. ■

As for ARCH(1), $\nu_t = y_t^2 - \sigma_t^2$ is a martingale difference (white noise when $Ey_t^4 < \infty$) and

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_q y_{t-q}^2 + \nu_t,$$

so $\{y_t^2\}$ is AR(q). Choosing the order q of an ARCH model therefore reduces to choosing the order of an AR model for $\{y_t^2\}$, using the model-selection tools of Chapter 5.

6.3 GARCH models

Motivation. A large ARCH order q is needed to describe persistent volatility, and then estimation is poor and the constraints are hard to satisfy. Adding lagged conditional variances, as ARMA adds lagged noise to AR, gives the parsimonious GARCH family.

Definition 6.3.1 (GARCH(1,1) model)

If $\{y_t\}$ satisfies

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \alpha_0 > 0,\ \alpha_1, \beta_1 \geq 0, \tag{6.12}$$

then $\{y_t\}$ follows a *GARCH(1,1) model*. If $\alpha_1 + \beta_1 < 1$, $\{y_t\}$ is (second-order) stationary; and if

$$E \log(\beta_1 + \alpha_1 \xi_t^2) < 0,$$

then $\{(y_t, \sigma_t)\}$ is strictly stationary (Nelson, 1990; Bougerol and Picard, 1992).

Proof of the stationarity statements. [Supplement] Put $a(x) = \beta_1 + \alpha_1 x^2$, so $\sigma_t^2 = \alpha_0 + a(\xi_{t-1})\sigma_{t-1}^2$ and, by iteration,

$$\sigma_t^2 = \alpha_0 \Big[1 + \sum_{k=1}^{K} \prod_{i=1}^{k} a(\xi_{t-i})\Big] + \prod_{i=1}^{K+1} a(\xi_{t-i})\, \sigma_{t-K-1}^2. \tag{6.13}$$

Step 1: strict stationarity. Let $\gamma = E \log a(\xi_t) < 0$ (well defined, since $E \log a \leq \log Ea < \infty$; if $\gamma = -\infty$, replace $\log a$ by $\max(\log a, -M)$ with M large, which keeps the mean negative and only enlarges the products below). By the strong law of large numbers, $\frac{1}{k}\sum_{i=1}^{k} \log a(\xi_{t-i}) \to \gamma$ a.s., so $\left(\prod_{i=1}^{k} a(\xi_{t-i})\right)^{1/k} \to e^{\gamma} < 1$, and by the root test

$$\sigma_t^2 := \alpha_0 \Big[1 + \sum_{k=1}^{\infty} \prod_{i=1}^{k} a(\xi_{t-i})\Big]$$

converges a.s. Substituting shows $\alpha_0 + a(\xi_{t-1})\sigma_{t-1}^2 = \sigma_t^2$, so $y_t = \sigma_t \xi_t$ solves (6.12); (y_t, σ_t) is a fixed function of the shifted i.i.d. sequence, hence strictly stationary. It is the only strictly stationary solution with σ_t^2 a function of the past ξ's: for another one, $\tilde{\sigma}_t^2$, (6.13) gives $\tilde{\sigma}_t^2 - \sigma_t^2 = \prod_{i=1}^{K} a(\xi_{t-i})\,(\tilde{\sigma}_{t-K}^2 - \sigma_{t-K}^2)$; the product tends to 0 a.s., while the second factor has a fixed distribution (stationarity) and is therefore tight, so the right side tends to 0 in probability. The left side does not depend on K, hence it is 0.

Step 2: second-order stationarity. If $\alpha_1 + \beta_1 < 1$, Jensen's inequality gives $E \log a(\xi_t) \leq \log Ea(\xi_t) = \log(\alpha_1 + \beta_1) < 0$, so Step 1 applies. By Tonelli and independence, $E \prod_{i=1}^{k} a(\xi_{t-i}) = (\alpha_1 + \beta_1)^k$, hence

$$Ey_t^2 = E\sigma_t^2 = \alpha_0 \sum_{k \geq 0} (\alpha_1 + \beta_1)^k = \frac{\alpha_0}{1 - \alpha_1 - \beta_1} < \infty.$$

Exactly as in Proposition (2), $E(y_t \mid \mathcal{F}_{t-1}) = 0$, so $\{y_t\}$ is a stationary white noise. ∎

The two conditions differ. When $\alpha_1 + \beta_1 = 1$ and ξ_t^2 is nondegenerate, strict Jensen gives $E \log(\beta_1 + \alpha_1 \xi_t^2) < \log 1 = 0$: the series is strictly stationary although $Ey_t^2 = \infty$.

The parameters of a GARCH(1,1) model fitted to daily returns (for instance $r_t = 100 \log(P_{t+1}/P_t)$ of a stock index) typically have $\alpha_1 + \beta_1$ just below 1. The case $\alpha_1 + \beta_1 = 1$ is the *IGARCH model*

(integrated GARCH): a shock at one time keeps influencing all later variances. When $\alpha_1 + \beta_1$ is 1 or close to 1 the conditional variance is called *persistent*, for the following reason.

Proposition (Variance forecasts, [Supplement])

In the GARCH(1,1) model let $\phi = \alpha_1 + \beta_1$. For $k \geq 0$,

$$E(\sigma_{t+k}^2 \mid \mathcal{F}_{t-1}) = \alpha_0 \sum_{j=0}^{k-1} \phi^j + \phi^k \sigma_t^2, \tag{6.14}$$

which has the form $c + \phi^k \sigma_t^2$. If $\phi \approx 1$, the effect of today's variance on the forecast barely decays as k grows.

Proof. For $s \geq t$, since σ_s^2 is $\mathcal{F}_{s-1}$-measurable and $E(y_s^2 \mid \mathcal{F}_{s-1}) = \sigma_s^2$,

$$E(\sigma_{s+1}^2 \mid \mathcal{F}_{s-1}) = \alpha_0 + \alpha_1 E(y_s^2 \mid \mathcal{F}_{s-1}) + \beta_1 \sigma_s^2 = \alpha_0 + \phi \sigma_s^2.$$

For $k = 0$ the claim is trivial. If it holds for k, the tower property gives $E(\sigma_{t+k+1}^2 \mid \mathcal{F}_{t-1}) = E(E(\sigma_{t+k+1}^2 \mid \mathcal{F}_{t+k-1}) \mid \mathcal{F}_{t-1}) = \alpha_0 + \phi E(\sigma_{t+k}^2 \mid \mathcal{F}_{t-1}) = \alpha_0 \sum_{j=0}^{k} \phi^j + \phi^{k+1} \sigma_t^2$. ∎

6.3.1 Asymmetric models

Returns often react more strongly to bad news (negative returns) than to good news.

Definition (Threshold GARCH(1,1))

Let $x^+ = \max\{0, x\}$ and $x^- = \max\{-x, 0\}$. If $y_t = \sigma_t \xi_t$ with $\{\xi_t\} \sim IID(0,1)$ and

$$\sigma_t^2 = \alpha_0 + \alpha_{11}(y_{t-1}^+)^2 + \alpha_{12}(y_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2, \qquad \alpha_0 > 0,\ \alpha_{11}, \alpha_{12}, \beta_1 \geq 0,$$

then $\{y_t\}$ follows a *threshold* (or *asymmetric*) *GARCH(1,1) model*. Equivalently, $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \gamma_1 (y_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2$ with $\alpha_1 = \alpha_{11}$, $\gamma_1 = \alpha_{12} - \alpha_{11}$.

With $\alpha_{12} > \alpha_{11}$ a negative return raises the next variance more than a positive one of the same size, which is the typical pattern in stock returns.

6.3.2 Gaussian quasi-maximum likelihood

Assume for the moment $\xi_t \sim N(0,1)$, fix initial values y_0 and σ_0^2 (commonly $y_0^2 = y_1^2$, $\sigma_0^2 = y_1^2$), and compute recursively from $\theta = (\alpha_0, \alpha_1, \beta_1)^T$

$$\sigma_t^{2*}(\theta) = \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^{2*}(\theta), \qquad t = 1, \dots, n.$$

The conditional likelihood is $L(\theta) = \prod_{t=1}^n f(y_t \mid y_{t-1}, \dots)$ with

$$f(y_t \mid y_{t-1}, \dots) = (2\pi\sigma_t^{2*})^{-1/2} \exp\{-y_t^2/(2\sigma_t^{2*})\},$$

and the *Gaussian MLE* $\hat{\theta}$ maximizes

$$l(\theta) = \log L(\theta) = -\frac{n}{2} \log 2\pi - \frac{1}{2} \sum_{t=1}^n \Big\{ \log \sigma_t^{2*}(\theta) + \frac{y_t^2}{\sigma_t^{2*}(\theta)} \Big\}. \tag{6.15}$$

The same estimator is used when ξ_t is not normal; it is then called the *quasi-MLE* (QMLE), and it remains consistent and asymptotically normal.

Proposition (Asymptotic normality of the QMLE, [Supplement])

Let $\{y_t\}$ be a strictly stationary GARCH(1,1) series with true parameter θ_0 in the interior of a compact parameter set, $\kappa_\xi = E\xi_t^4 < \infty$, and suppose $\hat\theta \to \theta_0$ in probability. Let $D_t = \sigma_t^{-2}\,\partial\sigma_t^2/\partial\theta$ at θ_0 and

$$J = E\big(D_t D_t^T\big) = -2\,E\Big[\frac{\partial^2 \log f(y_t \mid y_{t-1},\dots)}{\partial\theta\,\partial\theta^T}\Big],$$

assumed positive definite. Then

$$\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N(0,\Gamma), \qquad \Gamma = (\kappa_\xi - 1)J^{-1}$$

(Francq and Zakoian, 2004). For normal ξ_t, $\kappa_\xi = 3$ and $\Gamma = 2J^{-1}$.

Proof. [Supplement] We argue at the level of the standard M-estimation scheme, taking for granted two technical facts proved by Francq and Zakoian: the initial values y_0, σ_0^2 have an asymptotically negligible effect (because $\sigma_t^{2*} - \sigma_t^2 = O(\beta_1^t)$), and D_t and the second derivatives of $\log\sigma_t^2$ have finite moments uniformly in a neighbourhood of θ_0. Write $l_t(\theta) = -\tfrac12\{\log\sigma_t^2 + y_t^2/\sigma_t^2\}$.

Step 1: score. $\partial l_t/\partial\theta = \tfrac12(y_t^2/\sigma_t^2 - 1)\sigma_t^{-2}\partial\sigma_t^2/\partial\theta$, which at θ_0 equals $s_t = \tfrac12(\xi_t^2 - 1)D_t$. Since D_t is $\mathcal{F}_{t-1}$-measurable and ξ_t is independent of $\mathcal{F}_{t-1}$, $\{s_t\}$ is a stationary ergodic martingale difference with $E(s_t s_t^T) = \frac{\kappa_\xi - 1}{4}J$. The martingale central limit theorem (Billingsley) gives $n^{-1/2}\sum_{t=1}^n s_t \xrightarrow{d} N\big(0, \frac{\kappa_\xi-1}{4}J\big)$.

Step 2: Hessian. Differentiating again,

$$\frac{\partial^2 l_t}{\partial\theta\,\partial\theta^T} = \frac12\Big(\frac{y_t^2}{\sigma_t^2} - 1\Big)\frac{\partial}{\partial\theta^T}\Big(\sigma_t^{-2}\frac{\partial\sigma_t^2}{\partial\theta}\Big) - \frac12\,\frac{y_t^2}{\sigma_t^2}\,\sigma_t^{-4}\frac{\partial\sigma_t^2}{\partial\theta}\frac{\partial\sigma_t^2}{\partial\theta^T}.$$

At θ_0, $y_t^2/\sigma_t^2 = \xi_t^2$ has conditional mean 1, so the expectation equals $-\tfrac12 J$. By the ergodic theorem and the uniform moment bound, $\frac1n\sum_t \partial^2 l_t(\bar\theta_n)/\partial\theta\partial\theta^T \to -\tfrac12 J$ in probability for every $\bar\theta_n \to \theta_0$.

Step 3: expansion. Since $\hat\theta$ is an interior maximizer for large n, a mean-value expansion of each coordinate of the score gives $0 = n^{-1/2}\sum_t s_t + \big(\frac1n\sum_t \partial^2 l_t(\bar\theta_n)\big)\sqrt{n}(\hat\theta - \theta_0)$ with $\bar\theta_n$ between $\hat\theta$ and θ_0. Hence $\sqrt{n}(\hat\theta - \theta_0) = 2J^{-1}n^{-1/2}\sum_t s_t + o_p(1)$, whose limit is normal with covariance $2J^{-1}\cdot\frac{\kappa_\xi-1}{4}J\cdot 2J^{-1} = (\kappa_\xi - 1)J^{-1}$. ∎

Example 6.3.2 (Gaussian MLE for ARCH(1))

Fix y_0 and let $y_1,\dots,y_n$ follow ARCH(1). The conditional log-likelihood is

$$l(\alpha_0,\alpha_1) = -\frac{n}{2}\log 2\pi - \frac12\sum_{t=1}^n\Big\{\log(\alpha_0 + \alpha_1 y_{t-1}^2) + \frac{y_t^2}{\alpha_0 + \alpha_1 y_{t-1}^2}\Big\},$$

and its maximizer is the MLE. The likelihood equations have no closed-form solution, so the estimate is computed numerically.

6.3.3 GARCH(p, q)

Definition 6.3.3 (GARCH(p, q) model)

If $\{y_t\}$ satisfies

$$y_t = \xi_t \sigma_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i y_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2, \quad \alpha_0 > 0,\ \alpha_i, \beta_j \geq 0, \tag{6.16}$$

then $\{y_t\}$ follows a *GARCH(p, q) model.*

Proposition (Stationarity and ARMA form of GARCH(p, q), [Supplement])

Let $S = \sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j$.
(1) If $S < 1$, (6.16) has a unique strictly stationary solution with σ_t^2 a function of $\xi_{t-1}, \xi_{t-2}, \ldots$ and $E\sigma_t^2 < \infty$; it satisfies $Ey_t^2 = E\sigma_t^2 = \alpha_0/(1-S)$, and $\{y_t\}$ and $\{\sigma_t^2\}$ are stationary (Bollerslev, 1986). Conversely, a solution with $E\sigma_t^2$ finite and constant in t forces $S < 1$.
(2) With $\nu_t = y_t^2 - \sigma_t^2$, a martingale difference (white noise if $Ey_t^4 < \infty$),

$$\Big(1 - \sum_{i=1}^{q} \alpha_i B^i - \sum_{j=1}^{p} \beta_j B^j\Big) y_t^2 = \alpha_0 + \nu_t - \sum_{j=1}^{p} \beta_j \nu_{t-j}, \tag{6.17}$$

so $\{y_t^2\}$ follows an ARMA$(\max\{p, q\}, p)$ model.

Proof.

Step 1: construction. Set $h_t^{(0)} = 0$ and $h_t^{(m+1)} = \alpha_0 + \sum_{i=1}^{q} \alpha_i \xi_{t-i}^2 h_{t-i}^{(m)} + \sum_{j=1}^{p} \beta_j h_{t-j}^{(m)}$. As in the proof for Definition 6.2.1, $h_t^{(m)} = g_m(\xi_{t-1}, \xi_{t-2}, \ldots) \geq 0$ is nondecreasing in m, ξ_{t-i} is independent of $h_{t-i}^{(m)}$, and $Eh^{(m+1)} = \alpha_0 + S\, Eh^{(m)}$, so $Eh^{(m)} \uparrow \alpha_0/(1-S)$. The monotone limit σ_t^2 has $E\sigma_t^2 = \alpha_0/(1-S)$, satisfies (6.16) with $y_t = \sigma_t \xi_t$, and is strictly stationary as a fixed function of the shifted i.i.d. sequence; $Ey_t^2 = E\sigma_t^2$ and $E(y_t \mid \mathcal{F}_{t-1}) = 0$ make $\{y_t\}$ white noise.

Step 2: uniqueness. If $\tilde{\sigma}_t^2$ is another such solution, $D_t = \sigma_t^2 - \tilde{\sigma}_t^2$ satisfies $D_t = \sum_i \alpha_i \xi_{t-i}^2 D_{t-i} + \sum_j \beta_j D_{t-j}$, so $E|D_0| \leq S\, E|D_0|$ and $D = 0$.

Step 3: necessity. If $E\sigma_t^2 = m < \infty$ for all t, taking expectations in (6.16) with $Ey_{t-i}^2 = E\sigma_{t-i}^2$ gives $m = \alpha_0 + Sm$; as $\alpha_0 > 0$ and $m > 0$, $S < 1$.

Step 4: ARMA form. $E(\nu_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 E(\xi_t^2 - 1) = 0$; white-noise properties follow as in Proposition (4). Substituting $\sigma_{t-k}^2 = y_{t-k}^2 - \nu_{t-k}$ $(k = 0, \ldots, p)$ into (6.16),

$$y_t^2 - \nu_t = \alpha_0 + \sum_{i=1}^{q} \alpha_i y_{t-i}^2 + \sum_{j=1}^{p} \beta_j (y_{t-j}^2 - \nu_{t-j}),$$

which rearranges to (6.17). The AR polynomial has degree $\max\{p, q\}$ and the MA polynomial degree p. ■

Hence p, q can be chosen as in Chapter 5: maximize the Gaussian log-likelihood for each candidate order and minimize AIC or BIC. Since most financial series are well described by GARCH(1,1), higher orders are rarely needed.

Remark 6.3.4 (threshold AR(1)–GARCH(1,1))

In financial applications the following combination of a conditional mean and an asymmetric conditional variance is especially useful:

$$y_t = \phi y_{t-1} + \epsilon_t, \qquad \epsilon_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1),$$

$$\sigma_t^2 = \alpha_0 + \alpha_{11}(\epsilon_{t-1}^+)^2 + \alpha_{12}(\epsilon_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2, \qquad \alpha_0 > 0,\ \alpha_{11}, \alpha_{12}, \beta_1 \geq 0.$$

Heavy tails of ξ_t can also be built in directly, for example with a standardized t-distribution (*t-GARCH*); the Gaussian QMLE is then less efficient. Deciding whether ξ_t is normal is therefore of practical importance (Section 6.4). Another widely used variant works on the log scale.

Definition 6.3.5 (EGARCH(1,1) model)

If $\{y_t\}$ satisfies $y_t = \sigma_t \xi_t$, $\{\xi_t\} \sim IID(0,1)$, and

$$\log \sigma_t^2 = \alpha_0 + \theta \frac{y_{t-1}}{\sigma_{t-1}} + \gamma\Big\{\Big|\frac{y_{t-1}}{\sigma_{t-1}}\Big| - E\Big|\frac{y_{t-1}}{\sigma_{t-1}}\Big|\Big\} + \beta \log \sigma_{t-1}^2,$$

where $\alpha_0, \theta, \gamma \in \mathbb{R}$ and $|\beta| < 1$, then $\{y_t\}$ follows an *EGARCH(1,1) model* (exponential GARCH).

Because the recursion is for $\log \sigma_t^2$, no sign restrictions on the parameters are needed, which removes a drawback of GARCH. When $\theta < 0$, bad news ($\xi_{t-1} < 0$) raises volatility more than good news. Stochastic volatility models, in which $\log \sigma_t^2$ carries its own noise, are a further alternative not treated here.

6.4 Testing normality

Motivation. Errors in ARCH/GARCH models are frequently far from normal, and then the Gaussian MLE loses much of its efficiency. The Jarque–Bera test detects non-normality through skewness and kurtosis.

Given an i.i.d. sample $\xi_1, \dots, \xi_n$, to test H_0: $\xi_1, \dots, \xi_n$ *are normal*, the *Jarque–Bera statistic* is

$$T_{JB} = n\Big(\frac{S_n^2}{6} + \frac{(K_n - 3)^2}{24}\Big), \qquad S_n = \frac{m_3}{m_2^{3/2}}, \quad K_n = \frac{m_4}{m_2^2}, \quad m_k = \frac{1}{n}\sum_{t=1}^{n}(\xi_t - \bar{\xi})^k, \tag{6.18}$$

the sample skewness S_n and sample kurtosis K_n. Under normality these are near 0 and 3, so T_{JB} is small.

Proposition (Null distribution of the Jarque–Bera statistic, [Supplement])

Under H_0, $T_{JB} \xrightarrow{d} \chi^2(2)$.

Proof. All quantities are location-scale invariant, so take $\xi_t \sim N(0,1)$. Let $\hat{\mu}_k = \frac{1}{n}\sum \xi_t^k$; recall $E\xi^2 = 1$, $E\xi^4 = 3$, $E\xi^6 = 15$, $E\xi^8 = 105$, and odd moments vanish.

Step 1: reduction. $\sqrt{n}\,\bar{\xi} = O_p(1)$ and $\bar{\xi} \to 0$, $\hat{\mu}_2 \to 1$, $\hat{\mu}_3 \to 0$ a.s. Expanding,

$$m_2 = \hat{\mu}_2 - \bar{\xi}^2, \quad m_3 = \hat{\mu}_3 - 3\bar{\xi}\hat{\mu}_2 + 2\bar{\xi}^3, \quad m_4 = \hat{\mu}_4 - 4\bar{\xi}\hat{\mu}_3 + 6\bar{\xi}^2\hat{\mu}_2 - 3\bar{\xi}^4.$$

Hence $\sqrt{n}\,m_3 = \sqrt{n}(\hat{\mu}_3 - 3\bar{\xi}) + o_p(1)$, $\sqrt{n}(m_4 - 3) = \sqrt{n}(\hat{\mu}_4 - 3) + o_p(1)$ and $\sqrt{n}(m_2 - 1) = \sqrt{n}(\hat{\mu}_2 - 1) + o_p(1)$.

Step 2: delta method. Since $m_2 \to 1$, $\sqrt{n}S_n = \sqrt{n}\, m_3 + o_p(1)$. The map $(x, y) \mapsto x/y^2$ has gradient $(1, -6)$ at $(3, 1)$, so $\sqrt{n}(K_n - 3) = \sqrt{n}(m_4 - 3) - 6\sqrt{n}(m_2 - 1) + o_p(1)$. Therefore

$$\sqrt{n}\begin{pmatrix} S_n \\ K_n - 3 \end{pmatrix} = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} \begin{pmatrix} \xi_t^3 - 3\xi_t \\ \xi_t^4 - 6\xi_t^2 + 3 \end{pmatrix} + o_p(1).$$

Step 3: CLT. The summands are i.i.d. with mean 0 and $\mathrm{Var}(\xi^3 - 3\xi) = 15 - 18 + 9 = 6$, $\mathrm{Var}(\xi^4 - 6\xi^2) = 105 - 180 + 108 - 9 = 24$; their covariance is an odd moment, hence 0. By the multivariate CLT, $\sqrt{n}S_n$ and $\sqrt{n}(K_n - 3)$ are asymptotically independent $N(0, 6)$ and $N(0, 24)$, so $T_{JB} = (\sqrt{n}S_n)^2/6 + (\sqrt{n}(K_n - 3))^2/24 \xrightarrow{d} \chi^2(2)$. ∎

In a GARCH(1,1) model the errors are unobserved, so residuals are used. With the Gaussian MLE $\hat{\alpha}_0, \hat{\alpha}_1, \hat{\beta}_1$, write (6.12) as $(1 - \beta_1 B)\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$, i.e.

$$\sigma_t^2 = (1 - \beta_1 B)^{-1}\alpha_0 + \alpha_1(1 - \beta_1 B)^{-1} y_{t-1}^2 = \frac{\alpha_0}{1 - \beta_1} + \alpha_1 \sum_{j=1}^{\infty} \beta_1^{j-1} y_{t-j}^2,$$

and truncate at the first observation:

$$\hat{\xi}_t = \frac{y_t}{\hat{\sigma}_t}, \qquad \hat{\sigma}_t^2 = \frac{\hat{\alpha}_0}{1 - \hat{\beta}_1} + \hat{\alpha}_1 \sum_{j=1}^{t-1} \hat{\beta}_1^{j-1} y_{t-j}^2, \qquad 1 \le t \le n. \tag{6.19}$$

(Alternatively $\hat{\sigma}_t^2$ may be computed by the recursion (6.12) started from $y_0 = \sigma_0 = y_1$; the test result is the same asymptotically.) The residual statistic $\hat{T}_{JB}$, obtained by replacing ξ_t with $\hat{\xi}_t$ in (6.18), has the same $\chi^2(2)$ limit, and normality is rejected at level α if $\hat{T}_{JB} \ge c_\alpha$, where $P(\chi^2(2) \ge c_\alpha) = \alpha$.

6.5 Change-point detection

Motivation. GARCH parameters of real series often shift over time, and ignoring a shift misrepresents the model, typically as a spurious IGARCH with $\alpha_1 + \beta_1 \approx 1$. The CUSUM test compares partial sums of squared residuals with their expected linear growth.

Given $y_1, \dots, y_n$, test

> H_0: $y_1, \dots, y_n$ follow a GARCH(1,1) model with no parameter change;
> H_1: $y_1, \dots, y_n$ follow GARCH(1,1) but the parameters change at some time.

Under H_0 compute the Gaussian MLE, the residuals (6.19), and the *CUSUM statistic* of squared residuals

$$\hat{T}_n = \max_{m \le k \le n} \frac{1}{\sqrt{n}\,\hat{\tau}} \Big| \sum_{t=1}^{k} \hat{\xi}_t^2 - \frac{k}{n} \sum_{t=1}^{n} \hat{\xi}_t^2 \Big|, \qquad \hat{\tau}^2 = \frac{1}{n} \sum_{t=1}^{n} \hat{\xi}_t^4 - \Big(\frac{1}{n} \sum_{t=1}^{n} \hat{\xi}_t^2 \Big)^2, \tag{6.20}$$

where $\hat{\tau}^2$ estimates $\tau^2 = \mathrm{Var}(\xi_1^2)$ and m is a small fixed integer (say between 5 and 10). Under H_0, $\hat{\xi}_t^2 \approx \xi_t^2$, so

$$\hat{T}_n \approx T_n = \max_{m \le k \le n} \frac{1}{\sqrt{n}\,\hat{\tau}} \Big| \sum_{t=1}^{k} \xi_t^2 - \frac{k}{n} \sum_{t=1}^{n} \xi_t^2 \Big|. \tag{6.21}$$

Since $\sum_{t \le k} \xi_t^2 \approx k$ and $\frac{k}{n} \sum_{t \le n} \xi_t^2 \approx k$, T_n is small under H_0. If the parameters change at k^* with k^*/n away from 0 and 1, then $\frac{1}{n} \big| \sum_{t \le k^*} \hat{\xi}_t^2 - \frac{k^*}{n} \sum_{t \le n} \hat{\xi}_t^2 \big|$ converges to a positive constant, so $\hat{T}_n \to \infty$; large values of $\hat{T}_n$ reject H_0.

Proposition (Limit of the CUSUM statistic, [Supplement])

Let ξ_t be i.i.d. with $E\xi_t^2 = 1$, $E\xi_t^4 < \infty$, $\tau^2 = \mathrm{Var}(\xi_t^2) > 0$, and let $\hat\tau \to \tau$ in probability. Then

$$T_n \xrightarrow{d} T = \sup_{0\le u\le 1} |B^0(u)|,$$

where $B^0(u) = W(u) - uW(1)$ is a Brownian bridge and W a standard Brownian motion on $[0,1]$.

Proof.

Step 1. By Donsker's functional CLT, $W_n(u) = \frac{1}{\tau\sqrt{n}}\sum_{t\le \lfloor nu\rfloor}(\xi_t^2 - 1)$ converges in distribution to W in $D[0,1]$.

Step 2. The centring cancels: $\sum_{t\le k}\xi_t^2 - \frac{k}{n}\sum_{t\le n}\xi_t^2 = \sum_{t\le k}(\xi_t^2-1) - \frac{k}{n}\sum_{t\le n}(\xi_t^2-1)$. Hence, with τ in place of $\hat\tau$, the statistic equals $\max_{m\le k\le n}|W_n(k/n) - \frac{k}{n}W_n(1)|$. As W_n is a step function, this differs from $\sup_{0\le u\le 1}|W_n(u) - uW_n(1)|$ by at most $|W_n(1)|/n + \max_{k<m}|W_n(k/n)| + m|W_n(1)|/n$, and $\max_{k<m}|W_n(k/n)| \le \frac{1}{\tau\sqrt{n}}\sum_{t<m}|\xi_t^2 - 1| \to 0$ because m is fixed. So the difference is $o_p(1)$.

Step 3. The map $x \mapsto \sup_u |x(u) - ux(1)|$ is continuous at every continuous x for the sup-norm, and W has continuous paths, so by the continuous mapping theorem $\sup_u|W_n(u) - uW_n(1)| \xrightarrow{d} \sup_u|W(u) - uW(1)| = T$. Finally $\tau/\hat\tau \to 1$ and Slutsky's lemma give $T_n \xrightarrow{d} T$. ∎

The law of T is the Kolmogorov distribution, $P(T \le x) = 1 - 2\sum_{k\ge1}(-1)^{k-1}e^{-2k^2x^2}$, and $P(T \ge 1.358) = 0.05$. Thus H_0 is rejected at the 5% level when $\hat T_n \ge 1.358$, and the change point is estimated by

$$\hat k = \arg\max_{1\le k\le n}\hat T_n(k), \qquad \hat T_n(k) = \frac{1}{\sqrt{n}\,\hat\tau}\Big|\sum_{j=1}^{k}\hat\xi_j^2 - \frac{k}{n}\sum_{j=1}^{n}\hat\xi_j^2\Big|. \tag{6.22}$$

Several change points are found by binary segmentation: repeat the test on $y_1,\dots,y_{\hat k}$ and $y_{\hat k+1},\dots,y_n$ until no further change is detected. The CUSUM test is most powerful for changes near $n/2$ and weaker for changes near either end of the sample. Fitting a single GARCH(1,1) across a change typically yields $\hat\alpha_1 + \hat\beta_1$ very close to 1, so an apparent IGARCH fit should prompt a change-point test; for many stock-price series the test indeed finds changes.

6.6 Value-at-Risk

Motivation. Value-at-Risk is a low conditional quantile of the return distribution. In a GARCH model this conditional quantile is the volatility times a fixed quantile of the error, which can be estimated by quantile regression.

For a random variable X with distribution function F_X and $0 < \tau < 1$, the *τ-quantile* is

$$F_X^{-1}(\tau) = \min\{y : F_X(y) \ge \tau\}$$

(the minimum exists because F_X is right-continuous and nondecreasing).

Proposition (Two properties of quantiles, [Supplement])

(1) For $a > 0$ and $b \in \mathbb{R}$,

$$F_{aX+b}^{-1}(\tau) = aF_X^{-1}(\tau) + b. \tag{6.23}$$

(2) If $E|X| < \infty$ and $\rho_\tau(x) = x(\tau - I(x < 0))$ (the *check function*), then

$$F_X^{-1}(\tau) = \min\Big\{m : \ m \text{ minimizes } E\rho_\tau(X - m)\Big\}. \tag{6.24}$$

Proof.

Step 1. For every x, $F_{aX+b}(ax+b) = P(aX+b \le ax+b) = F_X(x)$. As $x \mapsto ax+b$ is an increasing bijection of $\mathbb{R}$,

$$\min\{y : F_{aX+b}(y) \ge \tau\} = \min\{ax+b : F_X(x) \ge \tau\} = a\min\{x : F_X(x) \ge \tau\} + b.$$

Step 2. For fixed x, $u \mapsto \rho_\tau(x-u)$ is Lipschitz with derivative $I(x < u) - \tau$ except at $u = x$, so $\rho_\tau(x-m') - \rho_\tau(x-m) = \int_m^{m'} (I(x<u) - \tau)du$ for $m < m'$. The integrand is bounded, so Fubini gives, with $g(m) = E\rho_\tau(X-m)$ (finite since $\rho_\tau(x) \le |x|$),

$$g(m') - g(m) = \int_m^{m'} (P(X < u) - \tau)du = \int_m^{m'} (F_X(u) - \tau)du,$$

the two integrands differing only at the countably many atoms of X. Let $q = F_X^{-1}(\tau)$. For $m < q$, $F_X(u) < \tau$ on $[m, q)$, so $g(q) - g(m) < 0$; for $m > q$, $F_X(u) \ge \tau$ on $[q, m]$, so $g(m) - g(q) \ge 0$. Thus q minimizes g and no smaller m does. ∎

By (6.24), from a sample $X_1, \dots, X_n$ the quantile is estimated by the minimizer of

$$\frac{1}{n}\sum_{t=1}^{n} \rho_\tau(X_t - m). \tag{6.25}$$

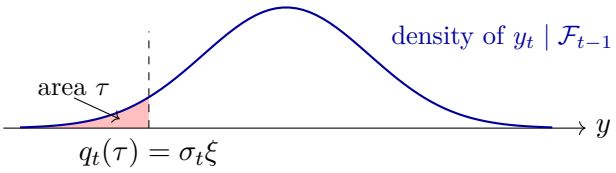


Figure 2: Conditional VaR at level τ: the τ-quantile of the conditional law of y_t, which scales with σ_t.

Now consider GARCH(1,1) in the reparametrized form

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim iid(0, \omega), \ \omega > 0, \qquad \sigma_t^2 = 1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \alpha, \beta \ge 0, \ \alpha + \beta < 1,$$

where the scale has been moved from the constant term into the error variance ω. Since σ_t is $\mathcal{F}_{t-1}$-measurable and ξ_t is independent of $\mathcal{F}_{t-1}$, (6.23) with $a = \sigma_t$, $b = 0$ gives the *conditional quantile*

$$q_t(\tau) = \sigma_t F_\xi^{-1}(\tau) = (1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2)^{1/2}\, \xi, \qquad \xi = F_\xi^{-1}(\tau).$$

Thus $q_t(\tau) = q_t(\xi, \alpha, \beta)$ depends on three parameters and, by the conditional version of (6.24),

$$q_t(\tau) = \min\{m : \ m \text{ minimizes } E(\rho_\tau(y_t - m) \mid \mathcal{F}_{t-1})\}.$$

Given $y_1, \dots, y_n$, the analogue of (6.25) is: choose $(\hat\xi, \hat\alpha, \hat\beta)$ to minimize

$$\frac{1}{n}\sum_{t=1}^{n} \rho_\tau(y_t - q_t^*(\xi, \alpha, \beta)), \qquad q_t^*(\xi, \alpha, \beta) = (1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^{2*})^{1/2}\xi,$$

where $\sigma_t^{2*} = 1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^{2*}$ is computed recursively from arbitrary initial values y_0, σ_0^{2*}. The estimate

$$\hat q_t(\tau) = q_t^*(\hat\xi, \hat\alpha, \hat\beta) =: \mathrm{VaR}_{Y_t \mid \mathcal{F}_{t-1}}(\tau)$$

is the *conditional Value-at-Risk* at level τ (Figure 2); in risk management τ is small, such as 0.01 or 0.05.

6.7 Selected exercises

Motivation. The exercises below fill in three steps that the ARCH(1) construction uses without proof: the conditional normal law, the uniqueness of the series solution, and the strict stationarity of functions of an i.i.d. sequence.

Exercise 6.1 (Conditional law of ARCH(1))

Let X, Y be independent random elements (with values in $\mathbb{R}^k$ or $\mathbb{R}^\infty$) and f a measurable real function. It is known that

$$P(f(X,Y) \le z \mid X = x) = P(f(x,Y) \le z).$$

Prove this identity, and use it to prove (6.3).

Solution.

Step 1: the identity. Fix z and put $g(x) = P(f(x,Y) \le z) = \int \mathbb{1}\{f(x,u) \le z\}\, P_Y(du)$. By independence the joint law of (X,Y) is the product $P_X \otimes P_Y$, and the integrand $\mathbb{1}\{f(x,u) \le z\}$ is jointly measurable; by Tonelli's theorem g is measurable and, for every Borel set A,

$$P(f(X,Y) \le z,\ X \in A) = \int_A \int \mathbb{1}\{f(x,u) \le z\}\, P_Y(du)\, P_X(dx) = E\big[g(X)\mathbb{1}\{X \in A\}\big].$$

This is the defining property of the conditional probability, so $g(x)$ is (a version of) $P(f(X,Y) \le z \mid X = x)$.

Step 2: application. By Proposition (1), y_{t-1} is a function of $\xi_{t-1}, \xi_{t-2}, \dots$ and hence independent of ξ_t. Take $X = y_{t-1}$, $Y = \xi_t$ and $f(x,u) = u\,(\alpha_0 + \alpha_1 x^2)^{1/2}$, so that $f(X,Y) = \sigma_t \xi_t = y_t$. Step 1 and $(\alpha_0 + \alpha_1 y^2)^{1/2} > 0$ give

$$P(y_t \le z \mid y_{t-1} = y) = P\big(\xi_t(\alpha_0 + \alpha_1 y^2)^{1/2} \le z\big) = \Phi\big(z/\sqrt{\alpha_0 + \alpha_1 y^2}\big),$$

which is (6.3).

Step 3: the whole past. The vector $X = (y_{t-1}, y_{t-2}, \dots)$ is also a function of $\xi_{t-1}, \xi_{t-2}, \dots$, hence independent of ξ_t. Applying Step 1 with this X and $f(x,u) = u(\alpha_0 + \alpha_1 x_1^2)^{1/2}$ shows that the conditional law of y_t given $\mathcal{F}_{t-1}$ is $N(0, \alpha_0 + \alpha_1 y_{t-1}^2) = N(0, \sigma_t^2)$. ■

Exercise 6.2 (Iterating the ARCH(1) recursion)

Consider the ARCH(1) model (6.2).
(a) Prove (6.4) for every $n \ge 0$.
(b) Let $\{y_t\}$ be a solution with $Ey_t^2 = c < \infty$ for all t, where each y_s is a function of $\xi_s, \xi_{s-1}, \dots$. Prove that y_t^2 and σ_t^2 are given by the series (6.5); consequently y_t equals (6.6).

Solution.

(a) Write $P_{t,j} = \xi_t^2 \xi_{t-1}^2 \cdots \xi_{t-j}^2$. For $n = 0$, $y_t^2 = \sigma_t^2 \xi_t^2 = (\alpha_0 + \alpha_1 y_{t-1}^2)\xi_t^2 = \alpha_0 P_{t,0} + \alpha_1 P_{t,0}\, y_{t-1}^2$. If (6.4) holds for n, substitute $y_{t-n-1}^2 = (\alpha_0 + \alpha_1 y_{t-n-2}^2)\xi_{t-n-1}^2$ into the remainder:

$$\alpha_1^{n+1} P_{t,n}\, y_{t-n-1}^2 = \alpha_0 \alpha_1^{n+1} P_{t,n+1} + \alpha_1^{n+2} P_{t,n+1}\, y_{t-n-2}^2,$$

which turns (6.4) for n into (6.4) for $n+1$.

(b)

Step 1. Let $S_n = \alpha_0 \sum_{j=0}^{n} \alpha_1^j P_{t,j}$ and $R_n = \alpha_1^{n+1} P_{t,n}\, y_{t-n-1}^2$, so $y_t^2 = S_n + R_n$ by (a). All terms are nonnegative, so S_n is nondecreasing and $R_n = y_t^2 - S_n$ is nonincreasing in n; hence $R_n \downarrow R$ a.s. for some $R \geq 0$.

Step 2. y_{t-n-1} is a function of $\xi_{t-n-1}, \xi_{t-n-2}, \ldots$, hence independent of $(\xi_t, \ldots, \xi_{t-n})$. Therefore

$$ER_n = \alpha_1^{n+1} E(\xi_t^2) \cdots E(\xi_{t-n}^2)\, E y_{t-n-1}^2 = \alpha_1^{n+1} c \to 0,$$

using $0 \leq \alpha_1 < 1$. Since $0 \leq R \leq R_n$, $ER = 0$, so $R = 0$ a.s. and $y_t^2 = \lim_n S_n = \alpha_0 \sum_{j \geq 0} \alpha_1^j P_{t,j}$, the first identity in (6.5).

Step 3. Applying Step 2 at time $t-1$ and using $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$,

$$\sigma_t^2 = \alpha_0 + \alpha_1 \alpha_0 \sum_{j \geq 0} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-1-j}^2,$$

the second identity in (6.5). As $\sigma_t \geq 0$ and $y_t = \sigma_t \xi_t$, y_t is given by (6.6). Together with Proposition (1) this shows that (6.6) is the *only* causal solution with finite variance. ■

Exercise 6.4 (Functions of an i.i.d. sequence)

Let $\{\xi_t\}$ be an i.i.d. sequence.
(a) For measurable $f_n : \mathbb{R}^n \to \mathbb{R}$, show that $Y_t^{(n)} = f_n(\xi_t, \ldots, \xi_{t-n+1})$ is strictly stationary (in particular for $n = 2$).
(b) Let $f : \mathbb{R}^\infty \to \mathbb{R}$ be measurable and suppose $f_n(\xi_t, \ldots, \xi_{t-n+1}) \to f(\xi_t, \xi_{t-1}, \ldots)$ a.s. for every t. Show that $Y_t = f(\xi_t, \xi_{t-1}, \ldots)$ is strictly stationary.

Solution.

(a) Fix times $t_1, \ldots, t_m$ and a shift h; let $L = \min_i t_i - n + 1$ and $U = \max_i t_i$. Each $Y_{t_i}^{(n)}$ is f_n applied to a fixed block of coordinates of $\boldsymbol{\xi} = (\xi_L, \ldots, \xi_U)$, so

$$(Y_{t_1}^{(n)}, \ldots, Y_{t_m}^{(n)}) = g(\boldsymbol{\xi}), \qquad (Y_{t_1+h}^{(n)}, \ldots, Y_{t_m+h}^{(n)}) = g(\boldsymbol{\xi}^h),$$

with the same measurable $g : \mathbb{R}^{U-L+1} \to \mathbb{R}^m$ and $\boldsymbol{\xi}^h = (\xi_{L+h}, \ldots, \xi_{U+h})$. By independence and identical distribution, both $\boldsymbol{\xi}$ and $\boldsymbol{\xi}^h$ have the product law $\mu^{\otimes(U-L+1)}$, where μ is the law of ξ_1. Measurable images of equally distributed vectors are equally distributed, so the two vectors of Y's have the same joint law, which is Definition 1.2.2. For $n = 2$, g consists of the coordinates $f_2(\xi_{t_i}, \xi_{t_i-1})$.

(b) Fix $t_1, \ldots, t_m$ and h. By assumption, $\mathbf{V}_n = (Y_{t_1}^{(n)}, \ldots, Y_{t_m}^{(n)}) \to \mathbf{V} = (Y_{t_1}, \ldots, Y_{t_m})$ a.s., and likewise $\mathbf{V}_n^h \to \mathbf{V}^h$ for the times shifted by h. Almost sure convergence implies convergence in distribution. By (a), $\mathbf{V}_n$ and $\mathbf{V}_n^h$ have the same law for every n, so this single sequence of laws converges weakly both to the law of $\mathbf{V}$ and to that of $\mathbf{V}^h$. Weak limits are unique, hence $\mathbf{V} \overset{d}{=} \mathbf{V}^h$, and $\{Y_t\}$ is strictly stationary. For example, with f_n the truncation of the series in (6.6) after its first $n-1$ terms (a function of $\xi_t, \ldots, \xi_{t-n+1}$), this proves the strict stationarity claimed in Proposition (1). ■